

Trekk App - Trekking Management Application

1. Student Details

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About Me: I am a Diploma Level Student at IIT Madras BS in Data Science and Applications program with a deep interest in Machine Learning and web-application development. I enjoy building meaningful applications that combine learning, analytics, and user experience.

2. Project Details

Project Title: Trekking Management Application

Problem Statement:

To build a **Trekking Management Application** web application that allows **Admin, Trek Staff, and Users (Trekkers)** to interact with the system based on their roles.

Approach:

The application was built using python-based Flask, which is a backend framework with modular structure. It allows users to browse Treks based on difficulty and location, track bookings and view the complete history of Trek Bookings and its profile. The Trek staff can manage assigned treks and its profile. The Admin can create, delete, update – user, trek staff and treks.

3. AI/LLM Declaration

I've used **Gemini (3.1 Pro)** to assist in writing SQLAlchemy model definitions, documentation samples, and improving style and responsiveness of the web-pages .

The extent of AI/LLM usage is around **10–15%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data

5. Database Schema / ER Diagram

Database Schema:



Tables:

1. **user**— stores user details (user_id, username, fullname, email, password, phone_number, role, status, is_validated, curr_status, qualification, date_of_create)
2. **trek** — stores trek details (trek_id, name, location,difficulty, duration, start_date,end_date, assigned_staff_id, price,total_slots, avl_slots, status, date_of_create, description)
3. **booking** — stores booking details (booking_id,user_id(FK),trek_id(FK), status, booking_date)

Relationships:

user:

- One-to-Many → user → trek(assigned_staff)
- One-to-Many → user → booking

trek:

- Many-to-One → trek → user(assigned_staff)
- One-to-Many → trek → booking

- Many-to-One → booking → user
- Many-to-One → booking → trek

6. Architecture and Features

Architecture Overview:

- **app.py** – main Flask application entry point
- **/models** – database models using SQLAlchemy
- **/routes** – Flask Blueprints for admin, user, staff and authentication
- **/templates** – Jinja2 HTML templates
- **/static** – CSS
- **/data.py** – Demo data generator for development and evaluation — not for production

Implemented Features:

1. User Authentication & Authorization:

- Secure login and registration with session management
- Password hashing using Werkzeug
- Role-based access control: Admin, Staff, User
- Custom login_required and role_required decorators per Blueprint

2. Admin Features:

- Dashboard with system-wide statistics
- Full trek lifecycle — create, edit, delete, assign staff, manage status
- Staff management — Validate staff based on – remove, blacklist / approve / pending

- User management — view and filter trekkers based on status —(blacklist / approved / pending)
- Global bookings overview with inline contextual search

3. Staff Features:

- Dedicated dashboard showing treks assigned to the logged-in staff member
- Real-time slot tracking and participant roster management
- Ability to update trek status (Mark as Started(Open), Mark as Closed(Closed), Mark as Completed(Completed).) and Available slots for the Trek
- Profile management — full name, phone, bio

4. User (Trekker) Features:

- Browse and explore treks with search and filtering by location and difficulty
- Single-click booking with real-time slot validation
- Active bookings view with cancellation support
- Permanent history of completed treks
- Profile management — full name, phone, bio

7.Video Presentation

Drive Link:

https://drive.google.com/file/d/1WJzL3Nm7a_jRVupVuxZWZ0Lw8y1bCGIV/view?usp=sharing